

Question 1

According to recent research from organizations such as the RAND Corporation and IBM, the failure rate for AI projects within organizations is as high as 70% to 80%. Taking into account the perspective of "Fear of Missing Out" (FOMO), please list and explain three key reasons why AI projects fail, and propose a specific improvement strategy for one of those reasons. (20%)

Question 2

Apple has officially announced a multi-year partnership to integrate Google's Gemini AI model into its Siri voice assistant. According to a joint statement, Apple concluded that Google's technology offers the most robust foundation for its "Apple Foundation Models," enabling a new generation of generative AI experiences for users. The partnership is intended to accelerate Apple's AI roadmap, which encountered setbacks and delays following its initial teaser at the 2024 Worldwide Developers Conference (WWDC). Analysts note that while reliance on an external partner underscores Apple's difficulties in developing a competitive in-house model, the agreement represents a strategic necessity. By leveraging Google's cloud computing and AI infrastructure, Apple aims to alleviate investor concerns about its perceived lag in the AI arms race and to stimulate iPhone demand through enhanced device capabilities.

Based on the case study, please address the following questions:

- (A). Apple's traditional sources of competitive advantage reside in ecosystem lock-in and superior user experience. Analyze how Apple's decision to adopt an external AI model (Google Gemini), rather than relying exclusively on in-house development, may help preserve or even strengthen its competitive position in the mobile device market. (10%)
- (B). Google functions simultaneously as a critical technology partner within Apple's iOS ecosystem and as the leader of the rival Android platform. Discuss the strategic risks (e.g., technological dependency) and potential benefits (e.g., reduced development time) associated with this co-opetitive relationship. (10%)
- (C). Apple has historically positioned user privacy as a core source of sustained competitive advantage. Given Google's data-driven business model, analyze how Apple should design its product architecture and/or communication strategy to ensure that its competitive advantage based on user trust is not undermined by this partnership. (10%)

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Question 3

Let X_1 and X_2 have the joint probability mass function

$$f(X_1, X_2) = \begin{cases} \frac{(|X_1| + |X_2| + |X_1 + X_2|)}{c}, & X_1 \in \{-1, 0, 1\}; X_2 \in \{0, 1\} \\ 0, & \text{otherwise} \end{cases}$$

- (A). Determine the normalizing constant c . (5%)
 (B). Find the joint probability mass function of $Y_1 = X_1 + X_2$ and $Y_2 = X_1 - X_2$. (5%)
 (C). Find the marginal probability mass function of Y_1 . (5%)

Question 4

Let Y_1 , Y_2 , and Y_3 be an independent and identically distributed random sample from a distribution with mean μ and variance $\sigma^2 < \infty$. Consider the following estimators for μ :

$$m_1 = Y_1, m_2 = \frac{Y_1 + \alpha Y_2}{1 + \alpha}, m_3 = \frac{Y_1 + 2Y_2 + \beta Y_3}{3 + \beta}, \text{ where } \alpha > 0 \text{ and } \beta > 0.$$

- (A). Determine which of the estimators m_1 , m_2 , and m_3 are unbiased for μ ? (10%)
 (B). Find the value of α that minimize $\text{Var}(m_2)$. (5%)
 (C). Find the value of β that minimize $\text{Var}(m_3)$. (5%)

Question 5

The mobile game *AngryRabbit* team examines users' premium treasure purchase behavior related to cumulative effective playtime. Premium treasures offer certain advantages in the game and are valuable to users. Moreover, users are classified as new, intermediate, or experienced based on their cumulative effective playtime. The following data shows the number of users by premium status and experience level. (15%)

	Level: New	Level: Intermediate	Level: Experienced	Total
Treasure purchased	45	50	65	160
Treasure not purchased	5	20	15	40
Total	50	70	80	200

Use a 5% level of significance and test whether the probability of premium treasure purchasing is independent of the user's cumulative effective playtime.

$$(\chi_{0.05,2}^2 = 5.99, \chi_{0.025,2}^2 = 7.38, \chi_{0.05,6}^2 = 12.59, \chi_{0.025,6}^2 = 14.45)$$